

equivalently, $|x''| \in (E')'_n$. To this end, let (f_α) be an arbitrary decreasing net in $(E')^+$ with $f_\alpha \downarrow 0$. It is enough to show that $\langle |x''|, f_\alpha \rangle \downarrow 0$. For each α we have

$$\begin{aligned} \langle |x''|, f_\alpha \rangle &= \sup\{|\langle x'', g \rangle| : |g| \leq f_\alpha\} \leq \sup\{|\langle g, x \rangle| : |g| \leq f_\alpha, x \in A\} \\ &\leq \rho_A(f_\alpha) \end{aligned}$$

To finish the proof, we need only to prove that $\rho_A(f_\alpha) \downarrow 0$. Indeed, it may be assumed that there exists an element $f \in E'$ such that $0 \leq f_\alpha \leq f$ for all α . Let $\varepsilon > 0$ be fixed. By (*) we have $\rho_A(f_\alpha) \leq f_\alpha(x) + \varepsilon$ for all α . Note that $f_\alpha(x) \downarrow 0$. It follows that $\inf_\alpha \rho_A(f_\alpha) \leq \varepsilon$. Since $\varepsilon > 0$ is arbitrary, we have $\inf_\alpha \rho_A(f_\alpha) = 0$, as desired. $\square$

In a KB -space without the weak Dunford-Pettis property (e.g., ℓ_p , $1 < p < \infty$) almost Dunford-Pettis sets and relatively weakly compact sets can not coincide.

There is another topic included in Bouras' question:

(Q1) *Is every almost Dunford-Pettis set in a KB -space L -weakly compact?*

We are not able to answer it and have to leave it still open. However, we can make some comments on (Q1). In [3] Chen *et al.* introduced the class of almost limited sets in Banach lattices. A norm bounded subset A of a Banach lattice E is said to be an *almost limited set* if every disjoint, weak* null sequence (f_n) of E' converges uniformly to zero on A . Clearly, every almost limited set in E is an almost Dunford-Pettis set. Almost limited sets and L -weakly compact sets coincide in E if and only if the norm of E is order continuous ([3, Theorem 2.6(2)]). Therefore, (Q1) is the same as the following question:

(Q1') *Do almost Dunford-Pettis sets and almost limited sets coincide in a KB -space?*

Let us recall that a Banach lattice E has the positive Schur property if, and only if, every relatively weakly compact subset of E is L -weakly compact. We know that every Banach lattice with the positive Schur property is a KB -space with the weak Dunford-Pettis property. In 1994 Wnuk [5] left a question which is still open until now:

(Q2) *Does every KB -space with the weak Dunford-Pettis property have the positive Schur property?*

By the Theorem of the present paper, in a KB -space with the weak Dunford-Pettis property almost Dunford-Pettis sets and relatively weakly compact sets coincide. Hence, if the answer to (Q1) is positive, then the answer to (Q2) is likewise positive. In other words, if the answer to (Q2) were negative, then there would exist a KB -space where we can find an almost Dunford-Pettis set which is not L -weakly compact.